Before we dive into the types of open source CMPs, let's discuss some of their broad capabilities and how they work.

According to analysis by Market Research Future (MRF), the global CMP market was valued at US$ 8.18 billion in 2018 and is expected to reach US$ 26.77 billion by 2022, registering a CAGR of 18.4 per cent during the forecast period. The market growth is attributed to the rising need for enterprises to have greater control over IT spending, the surge in the adoption of heterogeneous and multi-modal IT service delivery environments, the rapid deployment of virtualised workloads, and improved operational efficiency. On the other hand, insufficient technical expertise and the rising security concerns for platforms developed in-house are some of the factors expected to hinder the market growth during the assessment period.

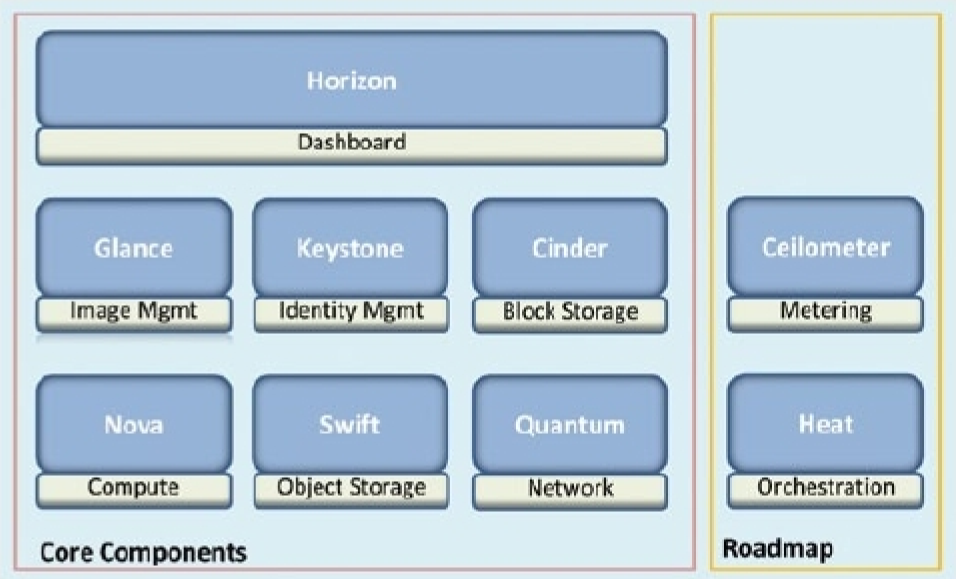

Fig. 2: OpenStack components

How cloud management platforms work

A CMP is deployed into existing cloud environments as a virtual machine (VM) consisting of a database and server. The server communicates with application programming interfaces (APIs) to connect the database and virtual resources held in the cloud. The database collects the information on how the virtual infrastructure is performing and sends an analysis to the Web interface, where systems administrators can analyse the cloud performance. The whole interconnectivity relies on the operating system, which commands all the different technologies that make up clouds and also deploys cloud management tools.

A CMP should be capable of the following things.

Strong integration with IT infrastructure. CMPs should be customised as per the enterprise's needs, and must meet the requirements of the operating systems, apps, storage frameworks and anything else running in the cloud.

Automating manual tasks. CMPs should have self-service capabilities to automate everything, with no human involvement.

Cost management. CMPs should assist organisations with precision cost forecasting and reporting to easily use and manage all sorts of cloud services.

Service management. They should assist the IT team to monitor cloud-based services to help in capacity planning, workload deployment, asset management and incident management.

Governance and security. CMPs should enable administrators to enforce policy-based control of cloud resources, and offer security features like encryption as well as identity and access management.

Top open source CMPs

The following are the top open source cloud management platform providers.

1. Apache CloudStack. Apache CloudStack is an open source, multi-hypervisor, multi-tenant, high-availability Infrastructure-as-a-Service CMP, which facilitates creating, deploying and managing cloud services by providing a complete stack of features and components for cloud environments. It uses existing hypervisors such as KVM, VMware vSphere, VMware ESXi, VMware vCenter and XenServer/XCP for virtualisation. CloudStack can also orchestrate the non-technical elements of service delivery such as billing and metering. It presents a range of APIs, allowing it to be integrated with any other platform.

The main features are:

- *Self-service user interface.* AJAX console access, multi-role support, network virtualisation, hypervisor agnostic, usage metering, virtual routers.
- *LVM support.* Block storage volumes, NetScaler support, OpenStack Swift integration, LDAP integration, domains and delegated administration.

Official website: *https://cloudstack.apache.org*

Latest version: 4.12.0.0

2. OpenStack. OpenStack consists of a set of software tools for building and managing cloud computing platforms for public and private clouds using pooled virtual resources. The tools comprising the OpenStack platform are called projects. They handle core cloud computing services of compute, networking, storage, identity and image services. OpenStack software controls large pools of compute, storage and networking resources throughout a data centre, and is managed through a dashboard or via the OpenStack API.

The main features are:

- *Services.* Messaging, clustering, containers, compute, identity, app data protection as a service, events, metadata indexing as service, workflows, DNS, database as a service, bare metal provisioning, optimisation and deployment, governance, and benchmarking.
- Web front-end, Big Data processing framework, container orchestration engine, key management, and NFV orchestration.

Official website: *http://www.openstack.org*

Latest version: Stein

3. ManageIQ. This is an open source CMP for hybrid IT environments, with a mix of public and private clouds. It provides tools for managing small and large environments as well as supports multiple technologies like virtual machines, public clouds and containers. It allows users to download any virtual appliance and deploy copies of it into virtualisation platforms like OpenStack or VMware. Three main variants of ManageIQ are available: Vagrant, Docker and Public Cloud.